Spatial Economics - TA Session 3

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Agenda

Three Approaches to Numerically Solve Systems of Equations

- Solvers
- Contraction Mappings
- Excess Demand

Solving Dynamic Spatial Models

Solvers

Newton-Raphson Method

- Start with a guess x_0
- 1st order Taylor approximation of f around x_n:

$$f(x) \approx f(x_n) + Df(x_n)(x - x_n)$$

- Find x_{n+1} that sets the approximation to zero since we are solving for $f(x) = 0$:

$$f(x_n) + Df(x_n)(x_{n+1} - x_n) = 0$$

- Rearrange to get the update rule:

$$x_{n+1} = x_n - [Df(x_n)]^{-1} f(x_n)$$

- **Strengths:**
  - Quadratic convergence near the root — very fast for smooth problems
- **Weaknesses:**
  - Must compute and invert the Jacobian at every step $\leftarrow$ Automatic differentiation (JAX, ForwardDiff.jl) lowers the computational burden
  - Sensitive to initial conditions — may diverge far from solution

Broyden's Method (Quasi-Newton)

Idea: Approximate the Jacobian via secant updates instead of recomputing it

- Same update as Newton, but replace $Df(x_n)$ with approximation B_n:

$$B_{n+1} = B_n + \frac{(\Delta f - B_n \Delta x)\,\Delta x^\top}{\Delta x^\top \Delta x}$$

  where $\Delta x = x_{n+1} - x_n$ and $\Delta f = f(x_{n+1}) - f(x_n)$

- **Strengths:**
  - No need to compute the Jacobian analytically
  - Cheaper per iteration than Newton-Raphson
- **Weaknesses:**
  - Superlinear but not quadratic convergence
  - Jacobian approximation can degrade over many iterations
  - Still sensitive to initial conditions

When to Use Off-the-Shelf Solvers

- Work well when the number of unknowns is **small**
- 200 countries $\Rightarrow$ 199 unknowns: performance degrades significantly
- Thousands of locations: not practical
- You still need to analytically prove existence and uniqueness

$\Rightarrow$ For spatial economics, we need methods that exploit **model structure**

Other Off-the-Shelf Methods

- **Levenberg-Marquardt:**
  - Hybrid of Newton-Raphson and gradient descent
  - Far from solution: cautious steps (like gradient descent)
  - Near solution: aggressive steps (like Newton)
  - More robust to initial conditions than pure Newton
- **Nelder-Mead (derivative-free):**
  - Uses only function evaluations — no Jacobian needed
  - Useful when f is noisy or non-differentiable
  - Slow convergence, but simple to implement

Contractions

Fixed Point

Definition

A **fixed point** of a function $f : X \to X$ is a point $x^* \in X$ such that

$$f(x^*) = x^*$$

- In economics: an equilibrium is often a fixed point of some operator
- Central question: does a fixed point **exist**? Is it **unique**? How do we **find** it?

A Selection of Fixed-Point Theorems

- **Brouwer's Theorem:** Every continuous function from a compact convex set to itself has a fixed point
- **Kakutani's Theorem:** Extends Brouwer to correspondences — the workhorse of equilibrium existence proofs
- **Tarski's Theorem:** Every monotone function on a complete lattice has a fixed point — used in supermodular games

Limitation: These theorems guarantee **existence**, but:

1. They say nothing about **uniqueness**
2. They provide no **algorithm** to find the fixed point

$\Rightarrow$ The **contraction mapping theorem** gives us all three: existence, uniqueness, and a computational algorithm

Contraction Mapping

Def: Contraction Mapping

A function $f : X \to X$ on a metric space (X, d) is a **contraction** if $\exists\, k \in (0, 1)$ such that

$$d\big(f(x), f(y)\big) \leq k \cdot d(x, y) \qquad \forall\, x, y \in X$$

- The constant k is the **contraction factor** (Lipschitz constant)
- Intuition: f brings any two points strictly closer together
- Smaller $k \Rightarrow$ faster convergence

Contraction Mapping Theorem (Banach Fixed-Point Theorem)

Theorem

If (X, d) is a **complete** metric space and $f : X \to X$ is a contraction with factor k, then:

1. There exists **exactly one** fixed point $x^* = f(x^*)$
2. For **any** initial $x_0 \in X$, the sequence $x_n = f(x_{n-1})$ converges to x^*
3. Rate of convergence:

$$d(x_n, x^*) \leq \frac{k^n}{1-k} d(x_1, x_0)$$

Showing contraction buys us **existence**, **uniqueness**, and a **computational algorithm**

A metric space is **complete** if every Cauchy sequence converges.

$\mathbb{R}^N$ with the Euclidean metric is complete — so this applies to our models.

Spectral Radius (Intuition with Linear Map f)

Definition

The **spectral radius** of a square matrix A is the largest eigenvalue in absolute value:

$$\rho(A) = \max_i |\sigma_i|$$

where $\sigma_1, \ldots, \sigma_N$ are the eigenvalues of A (possibly complex)

Intuition for its relevance when $f(x) = Ax + b$ is a linear map:

- Consider iterating $x_{n+1} = f(x_n)$ near a fixed point x^*
- Linearize (exact!): $x_{n+1} - x^* = Df\,(x_n - x^*)$ where $Df = A$ since f linear
- The error $\epsilon_n \equiv x_n - x^*$ evolves as $\epsilon_n = [Df]^n \epsilon_0$
- $[Df]^n \to 0$ if and only if $\rho(Df) < 1$

$\Rightarrow$ The spectral radius tells whether errors **shrink** or **diverge** under iteration

Spectral Radius (Non-linear f)

- But $f(x)$ is non-linear! Otherwise, we would have $x = f(x) = Ax + b = (I - A)^{-1}b$.
- When $f(x)$ is non-linear, linearization becomes a local approximation:
  - $x_{n+1} - x^* \approx Df(x^*)\,(x_n - x^*)$, i.e., $\epsilon_n \approx [Df(x^*)]^n\,\epsilon_0$
  - In this case, $\rho(Df(x^*)) < 1$ gives local convergence near x^*
- For global convergence,
  - one approach is checking $\sup_x \|Df(x)\| < 1$ (Banach via MVT) $\to$ too conservative
  - because $\|Df\|$ can overestimate $\rho(Df)$, where $\|\cdot\|$ is the induced matrix norm
  - Perov's theorem provides a sharper sufficient condition

Perov's Fixed-Point Theorem

Theorem (Perov, 1964)

Let $(\mathcal{X}, d)$ be a complete metric space with a **vector-valued metric** $d\colon \mathcal{X} \times \mathcal{X} \to \mathbb{R}_+^N$.
Let $f\colon \mathcal{X} \to \mathcal{X}$ satisfy, for each $i = 1, \ldots, N$:

$$d_i(f(x), f(y)) \le \sum_{j=1}^{N} A_{ij}\, d_j(x, y)$$

where A is a non-negative $N \times N$ matrix with $\rho(A) < 1$. Then:

1. f has a **unique** fixed point $x^* \in \mathcal{X}$
2. $f^{(k)}(x_0) \to x^*$ for any $x_0 \in \mathcal{X}$

Intuition:

- Distance is now a **vector**: track the error in each location separately
- A_{ij} encodes cross-location feedback and $\rho(A) < 1$ ensures the stability of feedbacks

Verifying Perov's Condition

- Crude approach earlier: check $\sup_x \|Df(x)\| < 1$ (collapses everything to a scalar)
- Sharper approach via Perov:
  1. Bound partial derivatives: $\sup_x |\partial f_i / \partial x_j| \leq k_{ij}$
  2. Stack bounds into matrix $K = [k_{ij}]$
  3. Check $\rho(K) < 1$
- Since $\rho(K) \leq \|K\|$ for any induced norm, whenever the norm check passes, Perov also passes, but not vice versa
- See Allen, Arkolakis, and Li (2024) for conditions in spatial models

Bonus: Blackwell's Sufficient Conditions

An easier-to-check criterion (Blackwell, 1965):

Blackwell's Sufficient Conditions

Let $T : B(X) \to B(X)$ be an operator on bounded functions. If:

1. **Monotonicity:** $f(x) \leq g(x)\ \forall x \implies (Tf)(x) \leq (Tg)(x)\ \forall x$
2. **Discounting:** $\exists\, \beta \in (0,1)$ such that $T(f + a)(x) \leq (Tf)(x) + \beta a$ for all $a \geq 0$

Then T is a contraction with modulus β.

- Originally developed for **Bellman operators** in dynamic programming
- Often easier to verify than the contraction property directly
- Stronger than $\sup_x \|Df(x)\| < 1$: Blackwell $\Rightarrow$ contraction in sup norm $\Rightarrow$ Banach
- The discount factor β in economic models naturally provides the discounting condition

Why We Use Contractions

- **Robust to initial conditions** — any starting point works
- **Guaranteed convergence** — no parameters to tune
- **Proves existence + uniqueness** alongside the algorithm
- **Scales to high dimensions** — $x_{n+1} = f(x_n)$ works the same regardless of dimension
- The workhorse of quantitative spatial economics

But: the equilibrium system does not always yield a contraction...

The Problem: When f Is Not a Contraction

- Sometimes the fixed-point iteration $x_{n+1} = f(x_n)$ has $\rho(Df) > 1$
- The iteration diverges or oscillates
- But we may know (or suspect) that a unique equilibrium exists
- Can we modify the iteration to make it converge?

Partial Updating with a Dampening Factor

Idea: Instead of $x_{n+1} = f(x_n)$, use a convex combination:

$$x_{n+1} = \lambda\, f(x_n) + (1-\lambda)\, x_n, \qquad \lambda \in (0,1)$$

- Define $g(x) = \lambda f(x) + (1-\lambda)x$. Its Jacobian is:

$$Dg = \lambda\, Df + (1-\lambda)\, I$$

- If σ_i are eigenvalues of Df, then eigenvalues of Dg are:

$$\mu_i = \lambda\, \sigma_i + (1-\lambda)$$

- For λ sufficiently small: $|\mu_i| < 1$ even when $|\sigma_i| > 1$, **provided** $\mathrm{Re}(\sigma_i) < 1$
- Dampening fixes oscillatory divergence ($\sigma < -1$), not monotonic divergence ($\sigma > 1$)
- When $\mathrm{Re}(\sigma_i) < 1$, dampening transforms a non-contraction into a contraction
- Trade-off: smaller $\lambda \Rightarrow$ more robust, but slower convergence

Fixed-Point Iteration: Cases

For real eigenvalues σ of Df at the fixed point:

Eigenvalue	**Behavior**	**Converges (w.o. dampening)**	**Method**
$\sigma \in (0, 1)$	Monotone	Yes	Direct iteration
$\sigma \in (-1, 0)$	Damped oscillation	Yes	Direct iteration
$\sigma < -1$	Explosive oscillation	No	Dampening
$\sigma > 1$	Monotonic divergence	No	

Anderson Acceleration

Idea: Speed up fixed-point iteration using recent iterations

- A quasi-Newton method applied to $r(x) = f(x) - x = 0$
- Standard iteration: $x_{n+1} = f(x_n)$ uses only the last iterate
- Anderson: store the last m residuals $r_k = f(x_k) - x_k$
- Find weights α_k that minimize $\left\|\sum_k \alpha_k r_k\right\|^2$
- Update using the optimal combination instead of just $f(x_n)$
- **In practice:**
  - $m = 3$ to 5 is typical
  - Can dramatically speed up convergent iterations
  - Used in computational trade and spatial models

Note: Does not turn a divergent iteration into a convergent one — use dampening first if needed, then accelerate with Anderson

Walras Tatonnement

Recall Walras's Law

- Let $Z(p)$ denote the excess demand vector across goods $l \in \{1, \ldots N\}$
- Equilibrium is characterized by $Z(p) = 0$

Walras's Law

In a competitive economy, the value of aggregate excess demand is zero:

$$p \cdot Z(p) = \sum_{l=1}^{N} p_l \, Z_l(p) = 0 \qquad \forall \, p$$

- An **accounting identity** from budget constraints
- Immediate: If $N - 1$ markets clear, the Nth clears automatically
- Reduces the dimensionality of the problem by one

Tâtonnement: Continuous-Time Analog

So far: discrete iteration $x_{n+1} = f(x_n)$, stability $\iff$ $\rho(Df) < 1$

Now: continuous dynamics, stability $\iff$ $\mathrm{Re}(\sigma_i) < 0$ for all eigenvalues

Tâtonnement Stability (Walras, 1874; Samuelson, 1947; Arrow-Block-Hurwicz, 1959)

Suppose the following hold.

1. $Z(p)$ satisfies Walras' law
2. There exists an equilibrium p^* with $Z(p^*) = 0$
3. Gross substitutes: $\frac{\partial Z_l}{\partial p_k} > 0$ for all $l \neq k$

Then, for all $c_l > 0$, the solution to the ODE

$$\dot{p}_l = c_l Z_l(p)$$

converges to a price vector proportional to p^*.

Excess Demand Method

Two ways to see convergence:

- **Spectral (local):** Linearize around p^*: $\dot{e} = C \cdot Dz(p^*)\, e$. Stable $\iff$ all eigenvalues have $\mathrm{Re}(\sigma) < 0$ where $e \equiv p - p^*$
- **Lyapunov (global):** Under gross substitutes, $p^* \cdot z(p) > 0$ for p not proportional to p^*. Then $V(p) = \|p - p^*\|^2$ is strictly decreasing along trajectories (MWG, Prop. 17.H.1)

In practice: discretize as $p_{n+1} = p_n + \alpha\, z(p_n)$, $\alpha > 0$

- This is **not** an optimization — no objective function being minimized, just a price adjustment process
- Choose α small enough to avoid oscillations and keep $p_n > 0$
- Gross substitutes is doing the work

Solving Dynamic Spatial Models

From Static to Dynamic

- In dynamic models, agents are **forward-looking**
- We must solve for entire **paths** that
  - clear all markets in each period
  - all intertemporal conditions are satisfied
- The challenge
  - expectations of future prices affect current decisions
  - current decisions effect future prices

Initial Value Problem (IVP) — Forward Iteration

- Initial Value Problem: solve for function $y(t)$ that satisfies

$$\text{Continuous time}: \quad \frac{dy(t)}{dt} = f(t, y(t)) \quad \text{given} \quad y(0) = y_0$$
$$\text{Discrete time}: \quad y_{t+1} = f(t, y_t) \quad \text{given} \quad y(0) = y_0$$

- Methods for solving numerically
  - Discrete time: just iterate forward
  - Continuous time: use finite-difference methods
- Finite-difference methods
  - Specify a grid for t: $t_0 < t_1 < \ldots < t_N$
  - Construct a difference equation that approximates the derivative

Boundary Value Problem (BVP) — Shooting

- BVPs impose conditions on the solution at multiple points, not only at $t = 0$
- Example: solve the system of ODEs

$$\begin{aligned} &\dot{x}(t) = f(x, y, t) \quad \text{and} \quad \dot{y}(t) = h(x, y, t) \\ &\text{where} \quad x(0) = x_0, \quad y(T) = y_T \end{aligned}$$

- We cannot iterate the system forward as we don't know $y(0)$
- We cannot iterate the system backward as we don't know $x(T)$
- Think of $y(T)$ as a condition for the new steady state
- Common method of numerically solving BVPs: shooting

Shooting Method: Idea

Take the previous example of BVP:

$$\dot{x}(t) = f(x, y, t) \quad \text{and} \quad \dot{y}(t) = h(x, y, t)$$
$$\text{where} \quad x(0) = x_0, \quad y(T) = y_T$$

Algorithm.

Set tolerance ϵ and make an initial guess for y_0. Call it $y_0^{(1)}$.

Starting from $i = 1$, iterate:

1. Using $(x_0, y_0^{(i)})$, solve the system forward to get $y^{(i)}(T)$
2. Check the convergence, $|y^{(i)}(T) - y_T| < \epsilon$
3. If not converged, update $y_0^{(i+1)}$ (How?)

Shooting Method: example from Judd (1998), chapter 10

Investment problem:

$$\max \int_0^\infty e^{-rt} \left(\pi(k) - I - \frac{\gamma}{2} I^2 \right) dt$$

$$\text{subject to} \quad \dot{k} = I \quad \text{and} \quad k(0) = k_0$$

Shooting Method: example from Judd (1998), chapter 10

Investment problem:

$$\max \int_0^\infty e^{-rt} \left(\pi(k) - I - \frac{\gamma}{2} I^2 \right) dt$$
$$\text{subject to} \quad \dot{k} = I \quad \text{and} \quad k(0) = k_0$$

ALGORITHM.

1. Set Hamiltonian and derive ODE for state $k(t)$ and co-state $\lambda(t)$
2. Solve for the steady state (k^*, λ^*)
3. Consider a similar problem with long but finite horizon T and boundary conditions $k(0) = k_0$ and $k(T) = k^*$
4. Shooting method: guess $\lambda(T)$ and use the time-reversed system to solve for $k(0)$ while holding $k(T) = k^*$. Iterate until $|k_0^i - k_0|$ is small.

Shooting Method: a remark

- Why we do not just iterate forward in this example?
- Think about the phase diagram in the neoclassical growth model
- If we are far from the saddle path, we can diverge from the steady state
- Hence, small changes in our guess of λ_0 would lead to large changes in $K(T)$
- When we iterate backward, we avoid this problem
- See Judd (1998) for more details

Boundary Value Problem (BVP) — Forward-backward iteration

- If $\dot{y}(x, y, t)$ is unstable forward, then pinning down $Y(T)$ by guessing y_0 can be hard
- Take the previous example of BVP:

$$\dot{x}(t) = f(x, y, t) \quad \text{and} \quad \dot{y}(t) = h(x, y, t)$$
$$\text{where} \quad x(0) = x_0, \quad y(T) = y_T$$

- Suppose $\dot{x}(t)$ (stocks/allocs) is stable forward, $\dot{y}(t)$ (prices/values) is stable backward
- Then, we can do better:
  - Guess a full path of $\{y_t^{(0)}\}_{t=0}^T$
  - Iterate $x(t)$ forward using $\{x(t), y_t^{(0)}\}_{t=0}^T$ and get $\{x_t^{(0)}\}_{t=0}^T$
  - Iterate $y(t)$ backward using $\{x_t^{(0)}, y(t)\}_{t=0}^T$ and get $\{y_t^{(0^*)}\}_{t=0}^T$
  - Set $\{y_t^{(1)}\}_{t=0}^T = \{y_t^{(0^*)}\}_{t=0}^T$ and continue iterating
- If the composite system is a contraction (e.g. due to gross substitutes for allocations forward, discounting for values backward), then algorithm is guaranteed to converge